

Agniveer Vayu Science (Group X) - 25 July 2022 - Memory Based Paper

Physics

Question 1

A thermometer works on the principle of

Options:

- A. Law of stable equilibrium
- B. Zeroth law of thermodynamics
- C. First law of thermodynamics
- D. Second law of thermodynamics

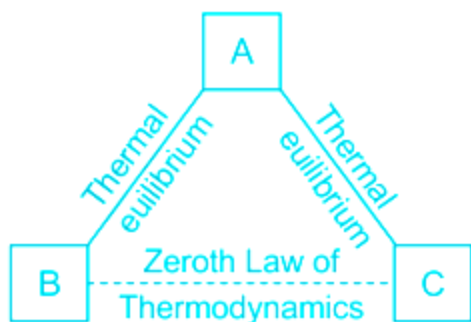
Answer: B

Solution:

Explanation:

The **zeroth law** of thermodynamics states that if two thermodynamic systems are each in thermal equilibrium with a third, then they are in thermal equilibrium with each other.

This law is the basis for the **temperature measurement**.



- By replacing the third body with a thermometer, the Zeroth law can be restated as two bodies are in thermal equilibrium if both have the same temperature reading even if they are not in contact.

- The thermometer is based on the principle of finding the temperature by measuring the thermometric property.
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Question 2

What are the dimensions of angular momentum?

Options:

- A. $[M L^2 T^{-2}]$
- B. $[M L^2 T^{-3}]$
- C. $[M L^2 T^{-1}]$
- D. $[M L T^{-1}]$

Answer: C

Solution:

CONCEPT:

- The dimensional formula is defined as the expression of the physical quantity in terms of mass, length, and time.
- The turning momentum of the particle about the axis of rotation is called the angular momentum of the particle. Mathematically angular momentum can be written as
- Either **Angular momentum** is the rotational analogue of linear momentum. The angular momentum of an object can also be connected to the angular velocity ω of the object via the moment of inertia I .

$$\therefore L = I \times \omega = r \times p$$

Where, L = angular momentum, I = moment of inertia, ω = angular momentum, r = distance and p = linear momentum = $(m \times v)$.

EXPLANATION:

Now,

Dimensional formula of $(r) = [L]$,

Dimensional formula of $(p) = [MLT^{-1}]$,

- Therefore, the dimensional formula of L is

$$\Rightarrow L = [L] \times [MLT^{-1}]$$

$$\Rightarrow L = [ML^2T^{-1}]$$

$\therefore$ Dimensional formula of angular momentum L is $[ML^2T^{-1}]$.



Important Point

- These are some of the useful dimensions of the basic physical quantity, Here M, L, T & I represent Mass, Length, Time, and Current respectively.

Physical quantity	Formula	Dimension
Velocity (v)	$v = dx/dt$	$[L^1T^{-1}]$
Angular velocity (ω)	$\omega = d\theta/dt$	$[T^{-1}]$
Acceleration (a)	$a = dv/dt$	$[L^1T^{-2}]$
Momentum (p)	$p = mv$	$[M^1L^1T^{-1}]$
Angular momentum (L)	$L = rp = mvr$	$[M^1L^2T^{-1}]$
Force (F)	$F = ma$	$[M^1L^1T^{-2}]$
Torque (τ)	$\tau = r \times F$	$[M^1L^2T^{-2}]$
Work (W)/Energy (E)	$W = Fx = \frac{1}{2}mv^2$	$[M^1L^2T^{-2}]$
Pressure (P)/stress	Pressure = Stress = F/Area	$[M^1L^{-1}T^{-2}]$
Electric current (I)	$I = dq/dt$	$[I^{-1}]$
Power	$P = \text{Work}/\text{time} = VI$	$[M^1L^2T^{-3}]$
Area	$A = x^2$	$[L^2]$

Question 3

Which of the colour of light is maximum deviated when passed through a prism?

Options:

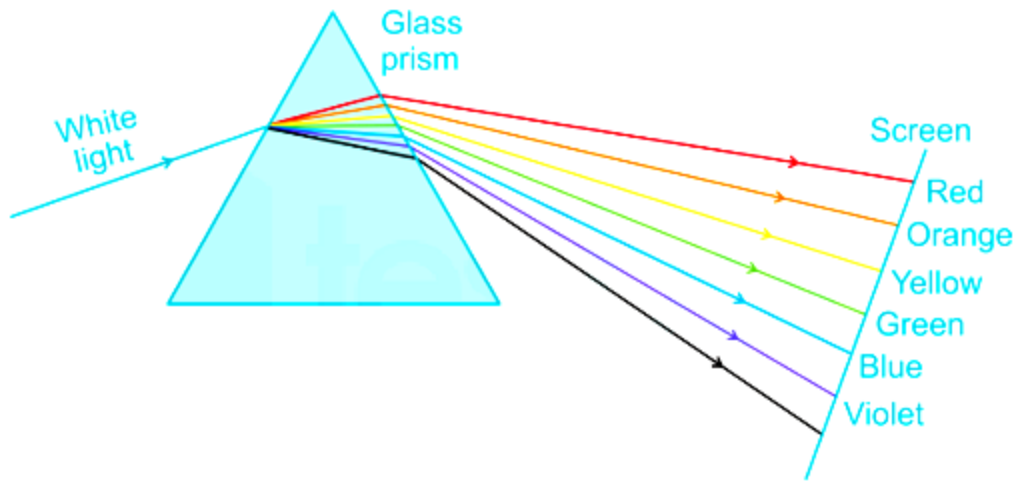
- A. Red
- B. Violet
- C. Blue
- D. Green

Answer: B

Solution:

CONCEPT:

Dispersion of light:



- Dispersion of light is the phenomenon of **splitting of a beam of white light into its constituent colours** on passing through a prism.
- The **band of seven colours** so obtained is called the **(visible) spectrum**.
- The **dispersion of white light** occurs because the **colours of white light travel at different speeds** through the glass prism.

EXPLANATION:

- When the **dispersion of white light** takes place through a prism, the **maximum deviation** is suffered by **violet colour** and **minimum deviation** is suffered by **red colour**. Therefore option 2 is correct.

Question 4

A boy runs from one end to other end of a semicircular track with radius 140 m. His displacement is:

Options:

- A. 280 m
- B. 180 m
- C. 140 m
- D. 493.6 m

Answer: A

Solution:

CONCEPT:

Distance:

- The measurement of how much any object moved is called **distance**.
- It is a **scalar quantity**.

Displacement:

- The **minimum distance between two points** is called the **displacement** of the object.
- It is a **vector quantity**.

CALCULATION:

Given:

Radius of semicircular track (r) = 140 m

- **Displacement of the semicircular track** is equal to the diameter of the circular track i.e.

$$\Rightarrow \text{Displacement} = 2R = 2 \times 140 = 280 \text{ m}$$



Important Point

- The **magnitude of displacement** is equal to the **minimum possible distance between two positions**. So **distance $\geq$ Displacement**.
 - For a moving particle, **distance can never be negative or zero** while **displacement can be (Zero displacement means that body after motion has come back to initial position)**. i.e., **Distance > 0 but Displacement $> = \text{or} < 0$** .
 - For motion between two points **displacement is single-valued** while **distance depends on the actual path** and so can have many values.
 - For a moving particle, **distance can never decrease with time** while **displacement can**.
 - A **decrease in displacement with time** means the **body is moving towards the initial position**.
-

Question 5

Which of the following is correct about the property of equipotential surface?

Options:

- A. The electric field is always parallel to the potential surface.
- B. An equipotential surface having variable voltage
- C. The electric field is always perpendicular to the potential surface.

D. None of the above

Answer: C

Solution:

Explanation:

Equipotential Surface

- An equipotential surface is a surface on which the potential, or voltage, is constant.
 - Electric field lines are always perpendicular to these surfaces, and the electric field points from surfaces of high potential to surfaces of low potential.
 - Suppose, for example, that a set of surfaces has been chosen so that their voltages are 5 V, 4 V, 3 V, 2 V, etc.
 - Then since the voltage difference between neighboring sheets is constant $\Delta V = 1 \text{ V}$, we can estimate the magnitude of the electric field between surfaces by the formula
 - $E = \frac{\Delta V}{l}$
 - Here l is the perpendicular distance between neighboring surfaces. (This formula is really just an approximate version of the path-integral definition of the voltage difference given above.)
 - Note that the electric field is strong where the equipotential surfaces are close together and weak where they are apart.
-

Question 6

The average radius of a soap bubble is 0.004 m. Find the excess pressure inside the soap bubble if the surface tension of soap bubble is 12 N/m.

Options:

- A. 0.048 Pa
- B. 0.050 Pa
- C. 12 KPa
- D. 5 KPa

Answer: C

Solution:

CONCEPT:

- **Surface tension (T)**- It is the property by virtue of which the free surface of a liquid at rest behaves like elastic stretched membrane tending to contract to occupy the minimum surface area.
- **Surface tension is the elastic tendency of a fluid surface** which makes it acquire the least surface area possible. Surface tension is measured in SI units of N/m (newton per meter).
- The **cohesive forces among liquid molecules** are responsible for the phenomenon of surface tension.

The **excess pressure** (P_{ex}) inside a **soap bubble** is given by:

$P_{ex} = 4 T/R$ where T is surface tension and R is the average radius of the soap bubble

EXPLANATION:

Given that: surface tension (T) = 12N/m

Average radius (R) = 0.004 m

$P_{ex} = 4 T/R = (4 \times 12)/0.004 = 12000 \text{ Pa} = 12 \text{ KPa}$



Additional Information

- As the soap bubble has two surfaces, so the **excess pressure in the soap bubble is two times that in the air bubble** or in a droplet.

Question 7

Angle between the electric component and magnetic component of an electromagnetic wave is ____.

Options:

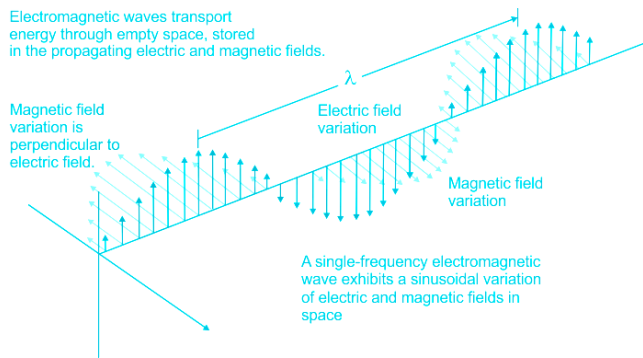
- A. 0
- B. π
- C. $\pi/2$
- D. $\pi/4$

Answer: C

Solution:

Explanation:

- The electric and magnetic field components of a linearly polarized electromagnetic wave oscillate in such a way that they peak at the same time and become zero at the same time, but they point to different directions in space, separated by an angle of 90° .



- Since there is no time difference between the peaks of the electric and magnetic oscillations, the phase difference between the electric and magnetic field vectors of a linearly polarized electromagnetic wave is zero.

Question 8

What is the current of an element in Ampere which produces 1200 Joules of heat in 6 seconds when the resistance of 12.5 ohms is applied to it?

Options:

- A. 8
- B. 4
- C. 2
- D. 1

Answer: B

Solution:

CONCEPT:

- **Heating effect of electric current:** When a current is flowing in a circuit having resistance there is heat dissipation due to the resistance. This is called the **heating effect of electric current**.

The **heat dissipated** is given by:

$$\text{Heat (H)} = I^2 R t$$

$$\text{Work done (W)} = \text{Heat (H)} = \text{Power (P)} \times \text{Time taken (t)}$$

Where I = the current flowing in the circuit, R = the resistance of the circuit, and t = the time taken

CALCULATION:

Given that:

H = heat generated (1200 J), R = Resistance = 12.5 ohms, t = time (6 s)

$$\text{Heat (H)} = I^2 R t$$

$$1200 = I^2 \times 12.5 \times 6$$

$$\text{Current, } I^2 = \frac{1200}{12.5 \times 6} = 16$$

$$(I) = 4 \text{ A}$$

Question 9

With the rise of temperature, the conductivity of a metals:

Options:

- A. decreases
- B. increases
- C. remains unchanged
- D. first decreases then increases

Answer: A

Solution:

CONCEPT:

- **Conductivity (k):** The property of materials due to which it allows the electric current to flow through them easily is called conductivity.
 - **Conductivity** is the reciprocal of **resistivity (ρ)**.

EXPLANATION:

- **Metals** conduct electricity due to the presence of **free electrons**.

- When heated, these **free electrons gain energy and collide with each other.**
- Due to this **heat the number of collisions between free electrons increases.**
- They **hinder the motion** of each other and hence overall **reduces conductivity.**
- Due to heat, there is **no change in the number of free electrons** in the metals.

Additional Information

- **Resistivity** increases with an increase in temperature.
 - Its reciprocal, conductivity decreases with temperature.
 - There **is** resistivity is given as $\rho(T) = \rho_0 (1 + \alpha \Delta T)$
 - Where $\rho(T)$ is resistivity at temperature T , ρ_0 is resistivity at some reference temperature.
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Question 10

Kepler's second law regarding the areal velocity of a planet is an application of the law of conservation of

Options:

- A. Angular momentum
- B. Energy
- C. Linear momentum
- D. None of these

Answer: A

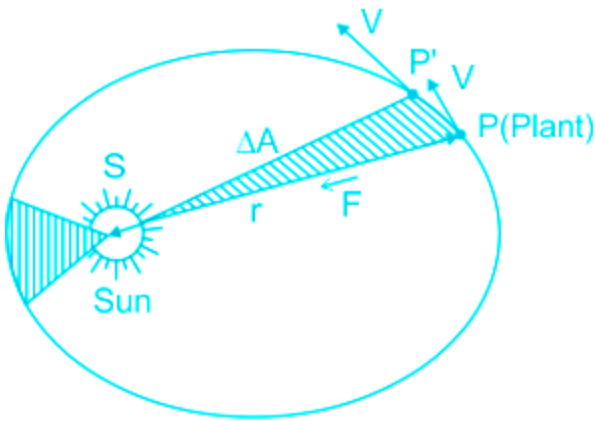
Solution:

CONCEPT:

Johannes Kepler proposed laws of planetary motion.

- **Kepler's Second Law:** The **line** that **joins** any planet to the sun **sweeps equal areas in equal intervals of time.**

This law comes from the observations that planets appear to **move slower** when they are **farther from the sun** than when they are **nearer**, they appear to **move faster**. It is also termed as the **Law of Equal Areas**.



The planet P moves around the sun in an elliptical orbit. The shaded area is the area ΔA swept out in a small interval of time Δt .

i. e. $\frac{dA}{dt} = \text{constant}$

- **Law of conservation of Angular Momentum:** The law of **conservation of angular momentum** states that when **no external torque acts** on an object, **no change of angular momentum** will occur.

EXPLANATION:

As we know from Kepler's second law

$$\frac{dA}{dt} = \text{constant}, \text{ and}$$

If take angular momentum 'L' and mass 'm' then it can be related by areal velocity is constant,

$$\frac{dA}{dt} = \frac{L}{2m} = \text{constant},$$

So this law can be derived from, the **law of conservation of angular momentum**. So option 1 follows.

So option 1 is correct.

Question 11

A bar magnet of magnetic moment M is placed in a magnetic field (B) at an angle of θ with the field. Find the work done in rotating the bar magnet by 180° .

Options:

A. MB

B. $0.5 MB$

C. $2 MB$

D. 4 MB

Answer: C

Solution:

CONCEPT:

- **Magnetic moment:** It is defined as the product of pole strength and the distance between the poles of a magnet.

$$\Rightarrow M = m \times 2l$$

where M = magnetic moment, m = pole strength and $2l$ = distance between the poles

- **Work done** in rotating a magnet of magnetic moment M with angle θ in an external magnetic field B is given as:

$$\Rightarrow W = MB (1 - \cos\theta)$$

EXPLANATION:

Since angle rotated (θ) = 180°

$$\text{Work done (W)} = MB (1 - \cos\theta) = MB (1 - \cos 180^\circ) = MB (1 - (-1)) = 2 MB$$

Question 12

If the cross-sectional area of a piece of metal is double, what will be its effect on its resistance (Keeping the length unchanged)?

Options:

- A. It will be doubled
- B. It will be halved
- C. It will quadruple
- D. It will not change

Answer: B

Solution:

The correct answer is It will be halved.

Key Points

CONCEPT:

- **Resistance:** The property of any conductor that opposes the flow of electric current through it and depends on the shape and size of the materials, temperature, and nature of the materials is called resistance.
 - It is denoted by R and the SI unit is the ohm (Ω).

The resistance is given by:

$$R = \rho \frac{L}{A}$$

where ρ is resistivity, L is the length and A is the area of the cross-section.

EXPLANATION:

Let the initial length and area of the cross-section be L and A respectively.

So resistance, $R = \rho \frac{L}{A}$

From the above, $R \propto \frac{1}{A}$

Given that:

The cross-sectional area is doubled, Final area (A') = 2A

New resistance, $R' = \rho \frac{L}{A'}$

From the above, $\frac{R'}{R} = \frac{A}{A'}$

$$\Rightarrow \frac{R'}{R} = \frac{A}{2A}$$

$$\Rightarrow R' = \frac{R}{2}$$

- Since the resistance is **inversely proportional to the cross-sectional area** of the metal, halving the area will double its resistance as it will make it two times more difficult for electrons to flow through the halved area.

Question 13

Which of the following statement is true regarding body is in equilibrium?

Options:

A. Body must be in translational equilibrium

B. Body must be in rotational equilibrium

C. Either 1 or 2

D. Both 1 and 2

Answer: D

Solution:

CONCEPT:

- **Equilibrium:** It is a **state** of a body in which **overall change** in a body is **zero**.
- **Translational Equilibrium:** A body is said to be in **translational equilibrium** if the **net sum of all the forces (F) acting on a body is zero**.

$$\Sigma F = 0$$

$$F_{\text{net}} = 0$$

$$F_1 + F_2 + F_3 + \dots + F_n = 0$$

- **Rotational Equilibrium:** A body is said to be in **rotational equilibrium** if the **net sum of all the torque (τ) acting on a body is zero**.

$$\Sigma \tau = 0$$

$$\tau_{\text{net}} = 0$$

$$\tau_1 + \tau_2 + \tau_3 + \dots + \tau_n = 0$$

EXPLANATION:

- A body is said to be in **equilibrium** if it is in translational equilibrium and **rotational equilibrium**.
- Both **translational and rotational equilibrium** is **required** to a body in equilibrium.

Hence **option 4** is correct.



Additional Information

- Solving the numerical problem on the equilibrium of forces always draws the **FBD (Free Body Diagram)**.
- A **simplified version** of the body **on which all external forces are shown** is called **FBD (Free Body Diagram)** of the given body.

Question 14

For two parallel wires, separation d carrying current I_a and I_b in the same direction will have the force on a length of L of either wire is

Options:

A. $\frac{(\mu_0 L I_a I_b)}{2\pi d}$ *Repulsion*

B. $\frac{\mu_0 L I_a I_b}{\pi d}$ *Attraction*

C. $\frac{(\mu_0 L I_a I_b)}{\pi d}$ *Repulsion*

D. $\frac{\mu_0 L I_a I_b}{2\pi d}$ *Attraction*

Answer: D

Solution:

Concept:

Magnetic field:

- The space or region around the current-carrying wire/moving electric charge or around a magnet in which force of magnetism can be experienced by other magnetic material is called a magnetic field/magnetic induction by that material/current.
- It is denoted by **B**.

The magnetic force per unit length between two parallel wires is given by;

$$F = \frac{\mu_0 I_1 I_2}{2\pi d}$$

Where μ_0 is the permeability of free space, I_1 is d.c current in the first wire and I_2 is d.c current in the second wire and d is the distance between two wires.

Explanation:

Given two wires carrying current I_a and I_b

The length of both wire is -

The force per unit length between the two wires is $\frac{F}{L} \Rightarrow F = \frac{\mu_0 L I_a I_b}{2\pi d}$

Since the current goes in **same direction** So it is of attraction.

Important Points

- If the conductor carries current in the same direction, then the force between them will be attractive.
 - If the conductor carries current in the opposite direction, then the force between them will be repulsive.
-

Question 15

A child is running towards a plane mirror with a velocity 5 m/s, then his image appears coming towards him with velocity:

Options:

A. 10 m/s

B. 2.5 m/s

C. 5m/s

D. 4m/s

Answer: A

Solution:

CONCEPT:

- **Plane Mirror:** A plane mirror is a mirror with a flat (planar) reflective surface.

The characteristics of an image formed in a plane mirror:

- The image formed by the plane mirror is **virtual and erect**, i.e. image cannot be projected or focused on a screen.
- The **distance of the image 'behind' the mirror** is the same as the **distance of the object in front of the mirror**.
- The **size of the image** formed is the same as the **size of the object**.
- The image is **laterally inverted**, i.e. left hand appears to be right hand when seen from the plane mirror.
- If the object moves **towards (or away from) the mirror at a certain rate**, the **image also moves towards (or away from) the mirror at the same rate**.

EXPLANATION:

Given that:

Speed of child toward the mirror (V) = 5 m/s.

- When the object moves with speed V towards (or away) from the plane mirror then image also moves towards (or away) with speed V .
- But the relative speed of image w.r.t. the object is $2V$.
- Therefore, relative speed of image w.r.t. object is $= 2 \times 5 = 10$ m/s.

Question 16

Choose the CORRECT statement regarding the time period of a simple pendulum.

Options:

- A. The time period of pendulum at pole of the earth is more than at equator.
- B. The time period of pendulum at pole of the earth is less than at equator.
- C. Time period is independent of acceleration due to gravity.
- D. All are correct.

Answer: B

Solution:

CONCEPT:

- The acceleration achieved by any object due to the gravitational force of attraction by any planet is called **acceleration due to gravity** by the earth.
- As each planet has a different mass and radius so the acceleration due to gravity will be different for different planets.

Acceleration due to gravity at any point is given by:

$$\Rightarrow g' = g - r \omega^2 \cos \varphi$$

Where g is the acceleration due to gravity on the earth surface, r is the radius at some random point, ω is angular acceleration and φ is the angle.

Acceleration due to gravity at any point is given by:

$$\Rightarrow g' = g - r \omega^2 \cos \varphi$$

On pole: $\varphi = 90^\circ$

$$\text{Acceleration due to gravity } (g'_p) = g - r \omega^2 \cos 90^\circ = g$$

On equator: $\varphi = 0^\circ$

$$\text{Acceleration due to gravity } (g''_e) = g - r \omega^2 \cos 0^\circ = g - r \omega^2$$

- **Simple pendulum:** An ideal simple pendulum consists of a **heavy point mass** (called **bob**) tied to one end of a **flexible, perfectly inextensible, and weightless string**.
- **The time period of a simple pendulum:**

$$T = 2\pi \sqrt{\frac{l}{g}}$$

Where T is the time period, l is the effective length of the pendulum, and g is the acceleration due to gravity.

EXPLANATION:

- From the above formula, the **time period is inversely proportional to the square of the acceleration due to gravity**. So option 3 is wrong.
 - As the **acceleration due to gravity at poles is more than that at the equator**. So the **time period of the pendulum at the pole of the earth is less than that at the equator**. So option 2 is correct.
-

Question 17

Which of the following equation shows the relationship between current (i) and drift velocity (V_d). (n is number of free electrons/ m^3 , A is the cross-sectional area, e is charge on an electron, R is resistance)

Options:

A. $V_d = I R^2$

B. $I = n e A V_d$

C. $V_d = I R$

D. $V_d = I^2 R t$

Answer: B

Solution:

CONCEPT:

- If the **charge is carried by the free electrons passing through a section of the wire in time t be Q** , then the strength of the current in the wire is

$$I = \frac{Q}{t}$$

- suppose, the **area of the cross-section of the wire is A** and the **number of free electrons per unit volume of the wire is n** .
- Then, the **number of electrons passing per second through a cross-section of the wire will be $n A V_d$** .
- Hence in t seconds, $n A V_d t$ electrons will pass. The **charge on an electron is e** . Therefore, the charge passing through any cross-section of the wire in t seconds will be

$$Q = (n A V_d t) \times e$$

- substituting the value of Q in equation (i).

$$I = neAV_d$$

- This is the **relation between electric current and drift velocity of free electrons.**

EXPLANATION:

$$I = neAV_d$$

- Hence **option 1 is correct.**

Additional Information

- The strength of the electric current in a conductor is measured by the magnitude of electric charge flowing per second through a cross-section of the conductor.
- suppose the ends of a metallic wire are connected to a battery so that an electric field E is established in the wire and the free electrons begin to move with drift velocity (V_d) in the direction opposite to the field (since electrons have a negative charge)

Question 18

What is the dimension of mutual Induction ?

Options:

A. $[ML^2T^{-3}A^{-1}]$

B. $[ML^2T^{-2}A^{-2}]$

C. $[ML^3T^{-4}A^{-1}]$

D. $[ML^3T^{-4}A^{-2}]$

Answer: B

Solution:

CONCEPT:

Mutual Induction:

- Whenever the **current passing through a coil or circuit changes**, the **magnetic flux** linked with a **neighboring coil or circuit** will also change.
- Hence a **emf will be induced in the neighboring coil or circuit**. This phenomenon is called '**mutual induction**'.

- **Mutual induction** between the two coils of area A, number of turns N_1 and N_2 with the length of secondary or primary l is given by:

$$\Rightarrow M = -\frac{e_2}{\frac{dI_1}{dt}} = -\frac{e_1}{\frac{dI_2}{dt}}$$

EXPLANATION:

- **Mutual induction** between the two coils of area A, number of turns N_1 and N_2 with the length of secondary or primary l is given by:

$$\Rightarrow M = -\frac{e_2}{\frac{dI_1}{dt}} = -\frac{e_1}{\frac{dI_2}{dt}}$$

- As we know, the **dimension of the electric potential** is

$$\Rightarrow e = [ML^2T^{-3}A^{-1}]$$

- The dimension of dI/dt is

$$\Rightarrow \frac{dI}{dt} = [AT^{-1}]$$

- Therefore, the **dimension of mutual Induction** is

$$\Rightarrow M = \frac{[ML^2T^{-3}A^{-1}]}{[AT^{-1}]} = [ML^2T^{-2}A^{-2}]$$



Additional Information

Self-Induction:

Whenever the electric current passing through a coil changes, the magnetic flux linked with it will also change.

- As a result of this, in accordance with **Faraday's laws of electromagnetic induction**, an emf is induced in the coil which opposes the change that causes it.
- This phenomenon is called '**self-induction**' and the emf induced is called **back emf**, current so produced in the coil is called **induced current**.
 - The **self-inductance of a solenoid** is given by:

$$\Rightarrow L = \frac{\mu_0 N^2 A}{l}$$

Question 19

What is the net increase or decrease in internal energy of the system, when 25J is transferred by heat into the system and 45J work is done by the system?

Options:

A. -70 J

B. 70 J

C. -20 J

D. 20 J

Answer: C

Solution:

CONCEPT:

- The **first law of thermodynamics** gives the relationship between heat, work, and internal energy.
- The **first law of thermodynamics** applies the conservation of energy principle to systems.
- The **first law of thermodynamics** states that the change in internal energy of a closed system equals the net heat transfer into the system minus the net work done by the system.

In equation form, the **first law of thermodynamics** is given by:

$$Q = \Delta U + W$$

where Q is the sum of all heat transfers into and out of the system, W is the net work done by the system and ΔU is the change in internal energy.

CALCULATION:

Given that:

Heat supplied to the system $Q = +25 \text{ J}$

Work done by the system $W = +45 \text{ J}$

By the **first law of thermodynamics** $Q = \Delta U + W$

$$25 = \Delta U + 45$$

$$\Delta U = -20 \text{ J}$$

So the correct answer is **option 3**.

Question 20

The efficiency of a Carnot engine operating with reservoir temperatures of 100°C and -23°C will be

Options:

A. 33

B. 67

C. 40

D. 80

Answer: A

Solution:

CONCEPT:

Carnot engine:

- The theoretical engine which works on the Carnot cycle is called a Carnot engine.
- It gives the maximum possible efficiency among all types of **heat engines**.

Heat source:

- The part of the Carnot engine which provides heat to the engine is called a heat source.
- The temperature of the source is maximum among all the parts.

Heat sink:

- The part of the Carnot engine in which an extra amount of heat is rejected by the engine is called a **heat sink**.
- The amount of work which is done by the engine is called as **work done**.
- The **efficiency (η)** of a **Carnot engine** is given by:

$$\Rightarrow \eta = 1 - \frac{T_C}{T_H} = \frac{\text{Work done}(W)}{Q_{in}} = \frac{Q_{in} - Q_R}{Q_{in}}$$

Where T_C is the temperature of the sink, T_H is the temperature of the source, W is work done by the engine, Q_{in} is the heat given to the engine/heat input and Q_R is heat rejected.

CALCULATION:

Given:

$$T_1 = 100^\circ\text{C} = 273 + 100 = 373 \text{ K}$$

$$T_2 = -23^\circ\text{C} = 273 - 23 = 250 \text{ K}$$

- The efficiency of the Carnot engine:

$$\Rightarrow \eta = \frac{W}{Q_1} = \frac{Q_1 - Q_2}{Q_1} = \frac{T_1 - T_2}{T_1} = \frac{373 - 250}{373} = \frac{123}{373} = 0.3297$$

- Therefore the **Efficiency of the Carnot cycle is 33%**

Question 21

The work done to move a charge of 10 C. Between two point in equipotential surface of 5 Volt is

Options:

A. 50 J

B. 5J

C. Zero

D. None of these

Answer: C

Solution:

Concept:

- Electric charge is acquired by the transfer of electrons.
- It is having an SI unit coulomb.
- Potential difference: Work done in bringing a unit positive charge from one point to another is called the potential difference. It is having a unit volt.
- The equipotential surface is the surface having the same potential.
- Potential difference = Work done / Time

$$Volt = \frac{Joule}{charge}$$

Explanation:

Charge $q = 10 \text{ C}$

Potential $V = 5 \text{ Volt}$

Work done = W

Work done = Charge $\times$ potential difference

But here the charges are at the same potential.

Work done is zero as there is no potential difference as charges move on the equipotential surface.

So the correct option is zero.

Question 22

The unit of energy in MKS system is

Options:

- A. Volt
- B. Erg
- C. Ohm
- D. Joule

Answer: D

Solution:

The correct answer is **Joule**.



Key Points

- **Joule is a unit of energy and is indicated by 'J' in MKS System.**
- It is expressed as the work required to produce one watt of power for one second.
- Joule is the unit of work/energy; in the International System of Units or (SI).



Additional Information

- Work done is said to be 1 Joule, when a force of one newton actually moves a body through a distance of one meter in the direction of applied force.
- Erg is the absolute unit of work or energy on the CGS system.
- The relation between joule and ergs is $1 \text{ Joule} = 10^7 \text{ ergs}$.
- **The MKS system of units** is a physical system of measurement that uses the meter, kilogram, and second (MKS) as base units.
- Other important systems are: CGS (centimeter, gram, and second) and FPS (Foot, Pound, Second)

Question 23

A spring of spring constant $5 \times 10^3 \text{ N/m}$ is stretched initially by 2 cm from the unstretched position. Then, the work required to stretch it further by another 8 cm.

Options:

- A. 1 J
- B. 25 J

C. 24 J

D. 30 J

Answer: C

Solution:

Concept:

Elastic Potential Energy (U):

- When a spring is stretched or compressed from its normal position ($x = 0$) by a small distance x , then a restoring force is produced in the spring to bring it to the normal position.
- According to Hooke's law, this restoring force is proportional to the displacement x and its direction is always opposite to the displacement.
- Mathematically Elastic Potential Energy or Work done can be written as,

$$W = \frac{1}{2} kx^2$$

Where k = spring constant

Calculation:

Given:

$$k = 5 \times 10^3 \text{ N/m}, x_1 = 2 \text{ cm} = 2 \times 10^{-2} \text{ m}, x_2 = 8 \text{ cm} = 8 \times 10^{-2} \text{ m}$$

Work done to stretch the spring by 2 cm from the mean position,

$$\Rightarrow W_1 = \frac{1}{2} kx_1^2$$

$$\Rightarrow W_1 = \frac{1}{2} \times 5 \times 10^3 \times (2 \times 10^{-2})^2$$

$$\Rightarrow W_1 = 1 \text{ J}$$

Work done to stretch the spring by 10 cm from the mean position,

$$\Rightarrow W_2 = \frac{1}{2} k(x_1 + x_2)^2$$

$$\Rightarrow W_2 = \frac{1}{2} \times (5 \times 10^3) \times [(2 \times 10^{-2}) + (8 \times 10^{-2})]^2$$

$$\Rightarrow W_2 = 25 \text{ J}$$

Net work done to stretch the spring from 2 cm to 10 cm is,

$$W_T = W_2 - W_1$$

$$W_T = 25 - 1 = 24 \text{ J}$$

Question 24

If the distance between the plates of a parallel plate capacitor is halved keeping other parameters unchanged. The the new capacitance will become-

Options:

- A. Two times of the initial capacitance
- B. One third of the initial capacitance
- C. Nine times of the initial capacitance
- D. Remains unchanged

Answer: A

Solution:

CONCEPT:

The capacitance of a capacitor (C):

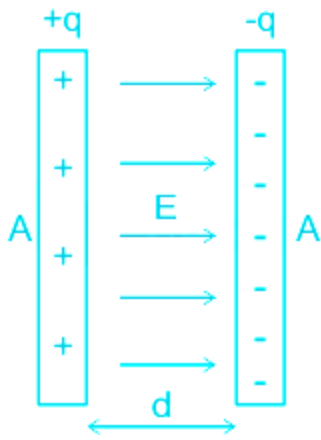
- The capacitance of a conductor is the ratio of **charge (Q)** to it by a rise in its **potential (V)**, i.e.

$$C = Q/V$$

- The unit of **capacitance** is the **farad**, (symbol **F**).
- A **parallel plate capacitor** consists of two large plane parallel conducting plates of area **A** and separated by a small distance **d**.
- Mathematical expression for the **capacitance of the parallel plate capacitor** is given by:
- $\Rightarrow C = \frac{\epsilon_0 A}{d}$

Where **C** = capacitance, **A** = area of the two plates, ϵ_0 = permittivity of free space, and **d** = separation between the plates,

Parallel Plate Capacitor:



EXPLANATION:

- The capacitance of the parallel plate capacitor is given by:

$$\Rightarrow C = \frac{\epsilon_0 A}{d}$$

- When the distance between the plates of a parallel plate capacitor is halved, the capacitance of the parallel plate capacitor will be

$$\Rightarrow C_1 = \frac{\epsilon_0 A}{\frac{d}{2}} = \frac{2\epsilon_0 A}{d} = 2C$$

- Hence, if the distance between the plates of a parallel plate capacitor is halved then the capacitance of the capacitor will become twice the initial capacitance.

Question 25

Above curie temperature a ferromagnetic substance

Options:

- A. becomes paramagnetic
- B. becomes diamagnetic
- C. becomes ferromagnetic
- D. All of the above

Answer: A

Solution:

Concept -

- **Ferromagnetic substance -**

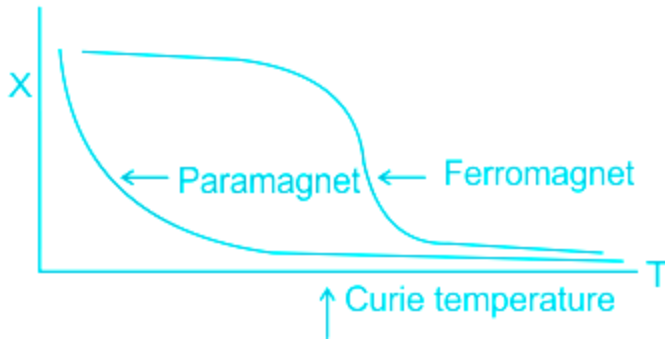
- Ferromagnetic materials have a spontaneous net magnetization at the atomic level, even when no external magnetic field is present.
- They are the materials that, when exposed to a magnetic field, tend to exhibit or display high magnetism in the direction of the field.
- The alignment patterns of these materials' component atoms are primarily responsible for their magnetic.
- They are **permanent magnets**.
- Examples: Iron, cobalt, nickel

- **Paramagnetic substance -**

- These are metals that have a **weak magnetic attraction**.
- The majority of electrons in these substances' atoms spin in the same direction but not all.
- Examples: Aluminium, Lithium, Magnesium

- **Diamagnetic substance -**

- Because the electrons in their atoms spin in equal amounts in opposite directions, diamagnetic substances have no observable reaction to the presence of a magnetic field.
- Diamagnetic materials, to put it simply, are compounds that are usually **repelled by a magnetic field**.
- Examples: Gold, Silver, Platinum.



- **Curie Temperature -**

- The temperature at which magnetic materials **lose their ferromagnetic characteristics** is known as the Curie temperature.
- Simply saying, it is the temperature at which certain materials lose their **permanent magnetic characteristics(ferromagnetism)** and are replaced by **induced magnetism(paramagnetism)**.
- Curie temperature is named after **Pierre Curie**, who discovered that magnetism vanishes at a certain temperature.
- At the curie temperature, hence, the **ferromagnetic material** loses its magnetism and becomes paramagnetic.

Explanation -

- At and above curie temperature, a substance starts to lose its ferromagnetic property.
- Thus, the permanent magnet moves toward being a paramagnetic substance.

Hence the correct answer to this question is - above curie temperature a **ferromagnetic substance becomes paramagnetic**

Question 26

The length of latus rectum of the parabola $x^2 = 20y$?

Options:

- A. 5
- B. 16
- C. 20
- D. 10

Answer: C

Solution:

Concept:

Parabola: The locus of a point which moves such that its distance from a fixed point is equal to its distance from a fixed straight line. (Eccentricity = $e = 1$)

Equation	$x^2 = 4ay$;
Vertex	(0, 0)
Focus	(0, a)
Equation of the directrix	$y = -a$
Equation of the axis	$x = 0$
Length of Latus rectum	4a
Focal distance	$y + a$

Calculation:

Given: $x^2 = 20y$

$$\Rightarrow x^2 = 4 \times 5 \times y$$

Compare with standard equation of parabola $x^2 = 4ay$

$$\text{So, } 4a = 4 \times 5$$

Therefore, Length of Latus rectum $= 4a = 4 \times 5 = 20$

Question 27

Evaluate, $\int \tan^3 x \sec^2 x \, dx$

Options:

A. $\sec^2 x + C$

B. $\frac{\tan^4 x}{4} + C$

C. $\frac{\tan^4 x}{2} + C$

D. $2 \tan x \sec x + C$

Answer: B

Solution:

Concept:

- $\frac{d \tan x}{dx} = \sec^2 x$

Calculation:

$$\int \tan^3 x \sec^2 x \, dx$$

Substitute $u = \tan x \Rightarrow \frac{du}{dx} = \sec^2 x$

$$du = \sec^2 x \, dx$$

$$= \int u^3 \, du$$

Apply power rule:

$$\int u^n \, du = \frac{u^{n+1}}{n+1} + C \text{ with } n = 3:$$

$$= \frac{u^4}{4} + C$$

After substitution $u = \tan x$

$$= \frac{\tan^4 x}{4} + C$$

Question 28

$|\bar{a}| = |\bar{b}| = |\bar{c}| = 1$ and $|\bar{a} + \bar{b} + \bar{c}|^2 = 0$, then find the value of $\bar{a}\bar{b} + \bar{b}\bar{c} + \bar{c}\bar{a}$.

Options:

A. $\frac{3}{2}$

B. $\frac{-3}{2}$

C. $\frac{1}{2}$

D. $\frac{-1}{2}$

Answer: B

Solution:

Solution:

$$|\bar{a} + \bar{b} + \bar{c}|^2 = 0$$

$$\Rightarrow |\bar{a}|^2 + |\bar{b}|^2 + |\bar{c}|^2 + 2(\bar{a}\bar{b} + \bar{b}\bar{c} + \bar{c}\bar{a}) = 0$$

$$\text{Given, } |\bar{a}| = |\bar{b}| = |\bar{c}| = 1$$

$$\Rightarrow 1^2 + 1^2 + 1^2 + 2(\bar{a}\bar{b} + \bar{b}\bar{c} + \bar{c}\bar{a}) = 0$$

$$\Rightarrow 3 + 2(\bar{a}\bar{b} + \bar{b}\bar{c} + \bar{c}\bar{a}) = 0$$

$$\Rightarrow 2(\bar{a}\bar{b} + \bar{b}\bar{c} + \bar{c}\bar{a}) = -3$$

$$\Rightarrow \bar{a}\bar{b} + \bar{b}\bar{c} + \bar{c}\bar{a} = -\frac{3}{2}$$

Question 29

What is the value of $(\frac{1+i}{1-i})$?

Options:

A. $0 + i$

B. $1 + i$

C. $0 + 2i$

D. $2 + 2i$

Answer: A

Solution:

Calculation:

$$\bullet \quad i^2 = -1$$

$$\left(\frac{1+i}{1-i} \right)$$

$$\Rightarrow \left(\frac{1+i}{1-i} \right) \times \left(\frac{1+i}{1+i} \right)$$

$$\Rightarrow \frac{(1+i)^2}{(1+i)(1-i)}$$

$$\Rightarrow \frac{1+i^2+2i}{1^2-i^2}$$

$$\Rightarrow \frac{1-1+2i}{1+1}$$

$$\Rightarrow \frac{2i}{2}$$

$$= i$$

Question 30

What is the probability that a die is thrown?

1) A even number

2) A odd number

3) A prime number

Options:

A. $\frac{1}{2}, \frac{1}{2}, \frac{1}{2}$

B. 1, 1, 1

C. $\frac{1}{4}, \frac{1}{4}, \frac{1}{4}$

D. q

Answer: A

Solution:

Solution:

We use the basic formula of probability to solve the problem.

The number of outcomes on throwing a die is $(1, 2, 3, 4, 5, 6) = 6$

(1) Total number of Even numbers are 2, 4 and 6 = 3

Probability of getting an even number = $\frac{\text{Number of Even Numbers}}{\text{Total Number of Outcomes}}$

$$\Rightarrow \frac{3}{6} = \frac{1}{2}$$

(2) Total number of Odd numbers are 1, 3 and 5 = 3

Probability of getting Odd number = $\frac{\text{Number of Odd Numbers}}{\text{Total Number of Outcomes}}$

$$\Rightarrow \frac{3}{6} = \frac{1}{2}$$

(3) Number of prime numbers on dice are 2, 3 and 5 = 3

Probability of getting a Prime number = $\frac{\text{Number of Prime Numbers}}{\text{Total Number of Outcomes}}$

$$\Rightarrow \frac{3}{6} = \frac{1}{2}$$

Question 31

From 5 males and 6 females, in how many ways can 5 be chosen to include at least one female?

Options:

A. 461

B. 455

C. 467

D. 500

Answer: A

Solution:

Explanation:

Number of ways to choose 5 members such that at least one female is included =

(Number of ways to choose 1 female and Number of ways to choose 4male) + (Number of ways to choose 2 female and Number of ways to choose 3male) + (Number of ways to choose 3 female and Number of ways to choose 2male) + (Number of ways to choose 4 female +Number of ways to choose 1male) + (Number of ways to choose 5female)

$$\Rightarrow (C_1^6 \times C_4^5) + (C_2^6 \times C_3^5) + (C_3^6 \times C_2^5) + (C_4^6 \times C_1^5) + (C_5^6) \text{ (taking possible combinations)}$$

$$\Rightarrow (6 \times 5) + (15 \times 10) + (20 \times 10) + (15 \times 5) + (6)$$

$$\Rightarrow 30 + 150 + 200 + 75 + 6$$

$$\Rightarrow 461$$

Number of ways to choose 5 members such that at least one female is included = 461.

Question 32

If the two points are $(a.\sin\alpha, a.\cos\alpha), (a.\sin\beta, a.\cos\beta)$, then what is the distance between them?

Options:

A. $2a \sin\left(\frac{\alpha-\beta}{2}\right)$

B. $a \sin\left(\frac{\alpha-\beta}{2}\right)$

C. $2a \cos\left(\frac{\alpha-\beta}{2}\right)$

D. $a \cos\left(\frac{\alpha-\beta}{2}\right)$

Answer: A

Solution:

Solution:

As per the distance formula:

$$\text{Distance between points } (x_1, y_1) \text{ and } (x_2, y_2) = \sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2}$$

Let, P(a sinα, a cosα) and Q(a sinβ, a cosβ)

Then the distance between P and Q is

$$PQ = \sqrt{(a \cdot \sin\alpha - a \cdot \sin\beta)^2 + (a \cdot \cos\alpha - a \cdot \cos\beta)^2}$$

$$PQ = \sqrt{a^2 \sin^2\alpha + a^2 \sin^2\beta - 2a^2 \sin\alpha \sin\beta + a^2 \cos^2\alpha + a^2 \cos^2\beta - 2a^2 \cos\alpha \cos\beta}$$

$$PQ = \sqrt{a^2(\sin^2\alpha + \cos^2\alpha) + a^2(\sin^2\beta + \cos^2\beta) - 2a^2(\sin\alpha \sin\beta + \cos\alpha \cos\beta)}$$

$$PQ = \sqrt{a^2 + a^2 - 2a^2(\sin\alpha \sin\beta + \cos\alpha \cos\beta)}$$

$$PQ = \sqrt{2a^2 - 2a^2(\sin\alpha \sin\beta + \cos\alpha \cos\beta)}$$

Using identity:

$$\cos(a - b) = \cos a \cdot \cos b + \sin a \cdot \sin b$$

$$PQ = \sqrt{2a^2 - 2a^2 \cos(\alpha - \beta)}$$

Using $\cos 2a = 1 - 2\sin^2 a$

$$PQ = \sqrt{2a^2 - 2a^2 \left(1 - 2\sin^2 \left(\frac{\alpha - \beta}{2}\right)\right)}$$

$$PQ = \sqrt{2a^2 - 2a^2 + 4a^2 \sin^2 \left(\frac{\alpha - \beta}{2}\right)}$$

$$PQ = \sqrt{4a^2 \sin^2 \left(\frac{\alpha - \beta}{2}\right)}$$

$$PQ = 2a \sin \left(\frac{\alpha - \beta}{2}\right)$$

Question 33

What is $\lim_{x \rightarrow 0} \frac{\sin x + \log(1-x)}{x^2}$ equal to?

Options:

A. 0

B. 1

C. -1/2

D. 1/2

Answer: C

Solution:

Concept:

L-Hospital Rule: Let $f(x)$ and $g(x)$ be two functions

Suppose that we have one of the following cases,

$$1. \lim_{x \rightarrow a} \frac{f(x)}{g(x)} = \frac{0}{0}$$

$$2. \lim_{x \rightarrow a} \frac{f(x)}{g(x)} = \frac{\infty}{\infty}$$

Then we can apply L-Hospital Rule $\Leftrightarrow \lim_{x \rightarrow a} \frac{f(x)}{g(x)} = \lim_{x \rightarrow a} \frac{f'(x)}{g'(x)}$

Note: We have to differentiate both the numerator and denominator with respect to x unless and until

$\lim_{x \rightarrow a} \frac{f(x)}{g(x)} = l \neq \frac{0}{0}$ where l is a finite value.

Calculation:

$$\lim_{x \rightarrow 0} \frac{\sin x + \log(1-x)}{x^2}$$

Form of limit $\frac{0}{0}$

Applying the L-Hospital rule,

$$\Rightarrow \lim_{x \rightarrow 0} \frac{\cos x + \frac{1}{1-x}(-1)}{2x}$$

$$\Rightarrow \lim_{x \rightarrow 0} \frac{\cos x - \frac{1}{1-x}}{2x}$$

$$\Rightarrow \lim_{x \rightarrow 0} \frac{\cos x - (1-x)^{-1}}{2x}$$

Now Again Form of limit $\frac{0}{0}$

Applying the L-Hospital rule,

$$\Rightarrow \lim_{x \rightarrow 0} \frac{-\sin x - \frac{(1-x)-(-1)}{(1-x)^2}}{2}$$

$$\Rightarrow \lim_{x \rightarrow 0} \frac{-\sin x - [(-1)(1-x)^{-2}(-1)]}{2}$$

$$\Rightarrow \lim_{x \rightarrow 0} \frac{-\sin x - [(1-x)^{-2}]}{2}$$

$$\Rightarrow \lim_{x \rightarrow 0} \frac{-\sin 0 - [(1-0)^{-2}]}{2}$$

$$= \frac{0-1}{2}$$
$$= -1/2$$

Question 34

Find the value of k for which the roots are real and equal in the equation $(k + 1)x^2 - 2(k - 1)x + 1 = 0$

Options:

- A. 3
- B. 0
- C. 5
- D. both a and b

Answer: D

Solution:

Solution:

The given equation is $(k + 1)x^2 - 2(k - 1)x + 1 = 0$

Comparing it with $ax^2 + bx + c = 0$, we get

$$a = k + 1, b = -2(k - 1) \text{ and } c = 1$$

$\therefore$ Discriminant,

$$D = b^2 - 4ac = 4(k - 1)^2 - 4(k + 1) \times 1$$

$$= 4(k^2 + 1 - 2k) - 4k - 4$$

$$= 4k^2 - 8k - 4k + 4 - 4$$

$$= 4k^2 - 12k$$

Since roots are real and equal, so

$$D = 0$$

$$\Rightarrow 4k^2 - 12k = 0$$

$$\Rightarrow 4k(k - 3) = 0$$

$$\Rightarrow 4k = 0 \text{ or } (k - 3) = 0$$

$$\Rightarrow \text{either } k = 0 \text{ or } k = 3$$

Hence, $k = 0, 3$

Question 35

The coefficient of x^4 in the expansion of $(x^3 + \frac{1}{x^2})^8$, is:

Options:

A.

50

B.

60

C.

70

D.

More than one of the above

Answer: C

Solution:

Concept:

The general term in the binomial expansion of $(x + y)^n$ is

$$T_{r+1} = {}^nC_r x^{n-r} y^r$$

Solution:

$$(x^3 + \frac{1}{x^2})^8$$

The general term in the binomial expansion is

$$T_{r+1} = {}^8C_r (x^3)^{8-r} \left(\frac{1}{x^2}\right)^r$$

$$T_{r+1} = {}^8C_r x^{24-3r} \frac{1}{x^{2r}}$$

$$T_{r+1} = {}^8C_r x^{(24-3r-2r)}$$

$$T_{r+1} = {}^8C_r x^{(24-5r)}$$

$$\text{Now, } 24 - 5r = 4$$

$$\Rightarrow 5r = 24 - 4$$

$$\Rightarrow 5r = 20$$

$$\Rightarrow r = 4$$

Hence, The coefficient of x^4 is 8C_4

$$= \frac{8!}{4!. (8-4)!}$$

$$= \frac{8!}{4!. (4)!}$$

$$= 70$$

Question 36

If 2 dice are thrown what is the probability of getting the same digits on both dice?

Options:

A. 1/3

B. 1/6

C. 1/36

D. 1/12

Answer: B

Solution:

Solution:

When two dice are thrown together, then the total number of all possible outcomes

$\{(1,1) (1,2) (1,3) (1,4) (1,5) (1,6)$

$(2,1) (2,2) (2,3) (2,4) (2,5) (2,6)$

$(3,1) (3,2) (3,3) (3,4) (3,5) (3,6)$

$(4,1) (4,2) (4,3) (4,4) (4,5) (4,6)$

$(5,1) (5,2) (5,3) (5,4) (5,5) (5,6)$

$(6,1) (6,2) (6,3) (6,4) (6,5) (6,6)\}$

$$\therefore n(S) = 6 \times 6 = 36$$

The favorable outcome of getting the same number on both dice is:

$(1, 1), (2, 2), (3, 3), (4, 4), (5, 5), (6, 6).$

$\therefore$ The number of favorable outcomes = 6

$$\therefore \text{Required probability} = \frac{6}{36}$$

$$= \frac{1}{6}$$

Question 37

If $y = 1 - x + \frac{x^2}{2!} - \frac{x^3}{3!} + \dots$ then what is the value of $\frac{dy}{dx}$?

Options:

A. e^{-x}

B. $-e^{-x}$

C. $-e^x$

D. e^x

Answer: B

Solution:

Concept:

Series Expansion of e^{-x} is

$$e^{-x} = 1 - x + \frac{x^2}{2!} - \frac{x^3}{3!} + \dots$$

Solution:

$$\text{Given, } y = 1 - x + \frac{x^2}{2!} - \frac{x^3}{3!} + \dots$$

$$y = e^{-x} = 1 - x + \frac{x^2}{2!} - \frac{x^3}{3!} + \dots$$

$$y = e^{-x}$$

$$\frac{dy}{dx} = \frac{d}{dx}(e^{-x})$$

$$\frac{dy}{dx} = (-1)e^{-x}$$

$$\frac{dy}{dx} = -e^{-x}$$

Question 38

Standard deviation of 1, 2, 3, 4, 5, 6, 7

Options:

A. 4

B. 7

C. 5

D. 2

Answer: D

Solution:

Concept:

Mean = Sum of observations/number of observations

$$\text{Variance} = \frac{1}{n} \sum (x_i - \bar{x})^2$$

Standard deviation = $\sqrt{\text{variance}}$

Calculation:

Given data: 1, 2, 3, 4, 5, 6, 7

Number of observation = $n = 7$

$$\text{Mean} = \frac{1+2+3+4+5+6+7}{7} = \frac{28}{7} = 4$$

$$\text{Now, Variance} = \frac{1}{n} \sum (x_i - \bar{x})^2$$

$$= \frac{1}{7} ((1 - 4)^2 + (2 - 4)^2 + (3 - 4)^2 + (4 - 4)^2 + (5 - 4)^2 + (6 - 4)^2 + (7 - 4)^2)$$

$$= \frac{1}{7} (9 + 4 + 1 + 0 + 1 + 4 + 9)$$

$$= 28/7$$

$$= 4$$

Now, standard deviation = $\sqrt{\text{variance}} = \sqrt{4} = 2$

Hence, option (4) is correct.

Question 39

Two letters are chosen from the letters of the word 'HOME'. The probability that both letters are vowels is

Options:

A. $\frac{1}{4}$

B. $\frac{1}{6}$

C. $\frac{2}{6}$

D. $\frac{2}{4}$

Answer: B

Solution:

Solution:

The total number of letters in the given word = 4, out of these O and E are two vowels.

Let S be the sample space, then

$$S = \{HO, HM, HE, OM, OE, ME\}$$

$$n(S) = 6$$

Let A be the event of taking two vowels, then

$$A = \{OE\} = \frac{n(A)}{n(S)}$$
$$= \frac{1}{6}$$

Question 40

The probabilities of solving a problem by three students A, B and C are $\frac{1}{2}$, $\frac{1}{3}$ and $\frac{1}{4}$, respectively. The probability that the problem will be not solved is

Options:

A. $\frac{1}{4}$

B. $\frac{1}{2}$

C. $\frac{3}{4}$

D. $\frac{1}{3}$

Answer: A

Solution:

Concept:

- Probability of event A or event B = $P(A \cup B)$
- Probability of event A and event B = $P(A \cap B)$
- If events A and B are independent events, then $P(A \cap B) = P(A) \times P(B)$
- Probability of not A = $P(\bar{A}) = 1 - P(A)$

Calculation:

The problem will be solved when either A solved the problem or B solved the problem or C solved the problem.

So, The probability that the problem will be solved = $P(A \cup B \cup C)$

Or the problem is not solved only when neither of A, B, and C solves the problem

$\Rightarrow$ A does not solve the problem and B does not solve the problem and C does not solve the problem,

$\Rightarrow$ The probability that the problem will be not solved = $P(\bar{A} \cap \bar{B} \cap \bar{C})$

$$\Rightarrow P(\overline{A}) \times P(\overline{B}) \times P(\overline{C})$$

$$\Rightarrow \text{The probability that the problem will be not solved} = (1 - P(A)) \times (1 - P(B)) \times (1 - P(C))$$

$$\Rightarrow \text{The probability that the problem will be not solved} = (1 - 1/2) \times (1 - 1/3) \times (1 - 1/4)$$

$$\Rightarrow \text{The probability that the problem will be not solved} = 1/2 \times 2/3 \times 3/4$$

$$\Rightarrow \text{The probability that the problem will be not solved} = 3/12 = 1/4$$

Question 41

Find the modulus of the complex number $\frac{1+2i}{1-(1-i)}$.

Options:

A. $\sqrt{5}$

B. $\sqrt{2}$

C. $\sqrt{7}$

D. 1

Answer: A

Solution:

Concept:

Modulus of the Complex Number $z = a + ib$ is given by:

$$\text{Mod}(z) = |z| = \sqrt{[\text{Re}(z)]^2 + [\text{Im}(z)]^2}$$

$$= \sqrt{a^2 + b^2}$$

Solution:

$$\frac{1+2i}{1-(1-i)} = \frac{1+2i}{i} = 2 + \frac{1}{i} = 2 + \frac{1}{i} \times \frac{i}{i} = 2 + \frac{i}{i^2} = 2 + \frac{i}{(-1)} = 2 - i$$

Modulus of the complex number $z = 2 - i$ is

$$\sqrt{2^2 + (-1)^2} = \sqrt{4 + 1} = \sqrt{5}$$

Question 42

Find the double derivative of $y = x^3$.

Options:

A. $3x$

B. $4x$

C. $5x$

D. $6x$

Answer: D

Solution:

Concept:

Differentiate using the Power Rule which states that

$$\frac{d}{dx}[x^n] = nx^{n-1}$$

Solution:

$$y = x^3$$

$$\frac{dy}{dx} = \frac{dx^3}{dx}$$

Using The Power Rule:

$$\frac{dy}{dx} = 3x^2$$

Since 3 is constant with respect to x , the derivate of $3x^2$ with respect to x is

$$3 \frac{d}{dx}[x^2]$$

$$\frac{d^2y}{dx^2} = 3 \frac{dx^2}{dx}$$

Using The Power Rule:

$$\frac{d^2y}{dx^2} = 3(2x)$$

$$\frac{d^2y}{dx^2} = 6x$$

Question 43

If $A = \begin{bmatrix} 1 & 3 & 3 \\ 1 & 4 & 3 \\ 1 & 3 & 4 \end{bmatrix}$ then $\text{adj}(A)$ is

Options:

A. $\begin{bmatrix} 7 & -3 & -3 \\ -1 & 1 & 0 \\ -1 & 0 & 1 \end{bmatrix}$

B. $\begin{bmatrix} 7 & 3 & 3 \\ -1 & 1 & 0 \\ -1 & 0 & -1 \end{bmatrix}$

C. $\begin{bmatrix} 7 & -3 & -3 \\ 1 & 1 & 0 \\ -1 & 0 & 1 \end{bmatrix}$

D. $\begin{bmatrix} -7 & 3 & -3 \\ -1 & 1 & 0 \\ -1 & 0 & -1 \end{bmatrix}$

Answer: A

Solution:

Concept:

Let $A = [a_{ij}]$ be a square matrix.

Steps for obtaining adjoint of the matrix

We need to calculate the cofactors of all elements

$$\Rightarrow C_{ij} \text{ or } A_{ij} = (-1)^{i+j} M_{ij}$$

Here, M_{ij} = Minor of matrix A

Calculation:

Given:

$$A = \begin{bmatrix} 1 & 3 & 3 \\ 1 & 4 & 3 \\ 1 & 3 & 4 \end{bmatrix}$$

$$\text{adj. } A = \begin{bmatrix} A_{11} & A_{21} & A_{31} \\ A_{12} & A_{22} & A_{32} \\ A_{13} & A_{23} & A_{33} \end{bmatrix}$$

$$\begin{bmatrix} 1 & 3 & 3 \\ 1 & 4 & 3 \\ 1 & 3 & 4 \end{bmatrix}$$

$$\Rightarrow A_{11} = 16 - 9 = 7$$

$$\Rightarrow A_{12} = -(4 - 3) = -1$$

$$\Rightarrow A_{13} = 3 - 4 = -1$$

$$\Rightarrow A_{21} = -(12 - 9) = -3$$

$$\Rightarrow A_{22} = 4 - 3 = 1$$

$$\Rightarrow A_{23} = -(3 - 3) = 0$$

$$\Rightarrow A_{31} = 9 - 12 = -3$$

$$\Rightarrow A_{32} = -(3 - 3) = 0$$

$$\Rightarrow A_{33} = 4 - 3 = 1$$

$$\Rightarrow \text{adj. } A = \begin{bmatrix} 7 & -3 & -3 \\ -1 & 1 & 0 \\ -1 & 0 & 1 \end{bmatrix}$$

$$\therefore \text{The value of adj. } A \text{ is } \begin{bmatrix} 7 & -3 & -3 \\ -1 & 1 & 0 \\ -1 & 0 & 1 \end{bmatrix}$$

Question 44

Evaluate: $(1 - \sec^2 x) \cot^2 x$

Options:

A. $\tan x$

B. $\sec x$

C. -1

D. 1

Answer: C

Solution:

Concept:

- $1 + \tan^2 x = \sec^2 x$
- $\tan x = \frac{1}{\cot x}$

Solution:

$$(1 - \sec^2 x) \cot^2 x$$

$$= [1 - (1 + \tan^2 x)] \cot^2 x$$

$$= -\tan^2 x \cdot \cot^2 x$$

$$= -\frac{1}{\cot^2 x} \cdot \cot^2 x$$

$$= -1$$

Question 45

Find the area bounded by the curve $y^2 = 4x$, the x axis and the lines $x = 4$ to 9 .

Options:

A. $\frac{19}{3}$

B. $\frac{76}{4}$

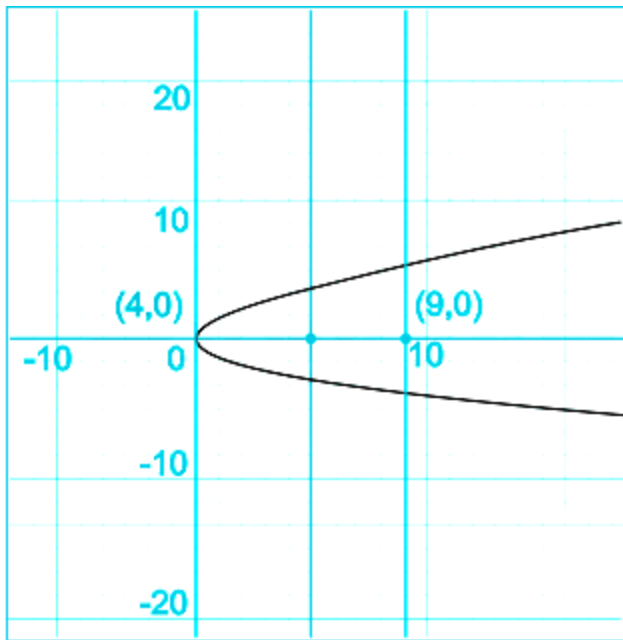
C. $\frac{4}{3}$

D. $\frac{76}{3}$

Answer: D

Solution:

Calculation:



The area bounded by the curves is $y^2 = 4x$,

$$\int_4^9 \sqrt{4x} dx$$

$$\Rightarrow 2 \int_4^9 \sqrt{x} dx$$

$$\Rightarrow 2 \left(\frac{x^{\frac{1}{2}+1}}{\frac{1}{2}+1} \right)_4^9$$

$$\Rightarrow 2 \left(\frac{x^{\frac{3}{2}}}{\frac{3}{2}} \right)_4^9$$

$$\Rightarrow 2 \times \frac{2}{3} \left(9^{\frac{3}{2}} - 4^{\frac{3}{2}} \right)$$

$$\Rightarrow \frac{4}{3} (3^3 - 2^3)$$

$$\Rightarrow \frac{4}{3} (27 - 8)$$

$$\Rightarrow \frac{4}{3} \times 19$$

$$\Rightarrow \frac{76}{3}$$

Question 46

Find the order and degree of the differential equation

$$\left(\frac{d^3y}{dx^3}\right)^2 = \sqrt{\frac{dy}{dx} + 1}?$$

Options:

A. 3, 4

B. 4, 3

C. 2, 1

D. 1, 2

Answer: A

Solution:

Concept:

Order of differential equation: The order of highest order derivative involved in a differential equation is called the order of the differential equation.

Degree of differential equation: The highest degree of the highest order derivative involved in a differential equation is said to be the degree of the differential equation. Before finding a degree in terms of differential coefficient it must be free from radicals, fractions and functions.

Calculation:

$$\left(\frac{d^3y}{dx^3}\right)^2 = \sqrt{\frac{dy}{dx} + 1}$$

$$\Rightarrow \left(\frac{d^3y}{dx^3}\right)^4 = \frac{dy}{dx} + 1$$

$$\Rightarrow \left(\frac{d^3y}{dx^3}\right)^4 - \frac{dy}{dx} - 1 = 0$$

Order = 3 and Degree = 4

Question 47

If $y = \tan x \cdot \tan 2x \cdot \tan 3x$, then find the value of $\frac{dy}{dx}$.

Options:

A. $3\sec^2 3x - 2\sec^2 2x$

B. $2\sec^2 2x - \sec^2 x$

C. $3\sin^2 3x - 2\sin^2 2x - \sin^2 x$

D. $3\sec^2 3x - 2\sec^2 2x - \sec^2 x$

Answer: D

Solution:

Concept:

$$\tan(a + b) = \frac{\tan a + \tan b}{1 - \tan a \cdot \tan b}$$

Explanation:

Given $y = \tan x \cdot \tan 2x \cdot \tan 3x$

As we know $\tan 3x = \tan(x + 2x) = \frac{\tan x + \tan 2x}{1 - \tan x \cdot \tan 2x}$

Thus, By Cross multiplication,

$$\tan 3x = \frac{\tan x + \tan 2x}{1 - \tan x \cdot \tan 2x}$$

we get,

$$\tan 3x (1 - \tan x \cdot \tan 2x) = \tan x + \tan 2x$$

$$\tan 3x - \tan x \cdot \tan 2x \cdot \tan 3x = \tan x + \tan 2x$$

$$\Rightarrow \tan x \cdot \tan 2x \cdot \tan 3x = \tan 3x - \tan 2x - \tan x$$

$$\Rightarrow \tan 3x - \tan 2x - \tan x = y$$

Differentiating with respect to x, we get

$$\frac{dy}{dx} = \frac{d(\tan 3x - \tan 2x - \tan x)}{dx}$$

$$\Rightarrow \frac{dy}{dx} = 3\sec^2 3x - 2\sec^2 2x - \sec^2 x$$

Question 48

If $\sin \theta + \cos \theta = m$ and $\sec \theta + \operatorname{cosec} \theta = n$, then find the value of $n(m + 1)(m - 1)$.

Options:

A. $n(m - 1)(m - 1) = 2m$

B. $n(m + 1)(m - 1) = 2m$

C. $n(m + 1)(m - 1) = m$

D. $n(m + 1)(m + 1) = 2m$

Answer: B

Solution:

Calculation:

$$m = \sin\theta + \cos\theta$$

$$\Rightarrow m^2 = (\sin\theta + \cos\theta)^2$$

$$\Rightarrow m^2 = \sin^2\theta + \cos^2\theta + 2\sin\theta.\cos\theta$$

$$\Rightarrow m^2 = 1 + 2\sin\theta\cos\theta$$

$$\Rightarrow \sin\theta\cos\theta = \frac{m^2-1}{2}$$

$$n = \sec\theta + \operatorname{cosec}\theta$$

$$\Rightarrow n = \frac{1}{\cos\theta} + \frac{1}{\sin\theta}$$

$$\Rightarrow n = \frac{\sin\theta + \cos\theta}{\sin\theta\cos\theta}$$

$$\Rightarrow n = \frac{m}{\frac{m^2-1}{2}}$$

$$\Rightarrow n(m^2 - 1) = 2m$$

$$\Rightarrow n(m + 1)(m - 1) = 2m$$

Question 49

If $x(x + y + z) = 9$, $y(x + y + z) = 16$, $z(x + y + z) = 144$, find the value of x .

Options:

A. 169

B. 0

C. $\frac{9}{13}$

D. 13

Answer: C

Solution:

Calculation:

$$x(x + y + z) = 9 \dots(1)$$

$$y(x + y + z) = 16 \dots(2)$$

$$z(x + y + z) = 144 \dots(3)$$

By adding these three equations

$$x(x + y + z) + y(x + y + z) + z(x + y + z) = 9 + 16 + 144$$

$$\Rightarrow (x + y + z)(x + y + z) = 169$$

$$\Rightarrow (x + y + z)^2 = 169$$

$$\Rightarrow (x + y + z) = 13$$

From eq. (1)

$$\Rightarrow x(x + y + z) = 9$$

$$\Rightarrow x(13) = 9$$

$$\Rightarrow x = \frac{9}{13}$$

Question 50

If $P(A \cup B) = 3/4$, $P(A \cap B) = 1/4$, $P(\bar{A}) = 2/3$, then find the value of $P(B)$.

Options:

A. $1/3$

- B. 1
- C. $\frac{5}{3}$
- D. $\frac{2}{3}$

Answer: D

Solution:

Concept:

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

Explanation:

$$P(A \cup B) = \frac{3}{4},$$

$$P(A \cap B) = \frac{1}{4},$$

$$P(\bar{A}) = \frac{2}{3}$$

$$\Rightarrow P(A) = 1 - \frac{2}{3} = \frac{1}{3}$$

$$\text{Since, } P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

$$\Rightarrow \frac{3}{4} = \frac{1}{3} + P(B) - \frac{1}{4}$$

$$\Rightarrow P(B) = \frac{2}{3}$$

English

Question 51

Select the most appropriate synonym of the following given word:

Abhorrent

Options:

- A. Innocuous
- B. Inoffensive

C. Attractive

D. Awful

Answer: D

Solution:

The correct answer is "**Awful**"

Key Points

- Let's see the meanings of the given words-
 - **Abhorrent**-causing or deserving strong dislike or hatred
 - **Example**-Racism of any kind is abhorrent to me.
 - **Innocuous**-completely harmless
 - **Example**-Some mushrooms look **innocuous** but are in fact poisonous.
 - **Inoffensive**-not causing any harm or offense.
 - **Example**-He seemed like a quiet, **inoffensive** sort of a guy.
 - **Attractive**-very pleasing in appearance or sound
 - **Example**-I find him very **attractive**
 - **Awful**-extremely bad or unpleasant
 - **Example**-He suffered **awful** injuries in the crash.
 - Hence, the correct answer is **Option 4 i.e. "Awful"**
-

Question 52

Out of the given four words only one is spelled correctly. Find the word that is correctly spelled.

Options:

A. Democratic

B. Suferred

C. Weaher

D. Horible

Answer: A

Solution:

The correct spelled word is "**Democratic**"

Key Points

- The correct spelling is "**Democratic**" which means "based on the principles of democracy"
 - Example-Do you think Australia is a more democratic country than Britain?
- Let's see the meanings of the other words-
 - **Suffered**-to experience physical or mental pain
 - **Weather**-the conditions in the air above the earth such as wind, rain, or temperature, especially at a particular time over a particular area
 - **Horrible**-very unpleasant or bad

Hence, the correct answer is **Option 1 i.e. Democratic.**

Question 53

Direction: Select the segment of the sentence that contains the grammatical error. If there is no error, mark 'No error' as your answer.

Hardly had he completed (A)/ his homework than the (B)/ lights went out. (C)/ No error (D)

Options:

- A. A
- B. B
- C. C
- D. D

Answer: B

Solution:

The correct answer is **B.**

Key Points

- A **conjunction** is a word that is used for **joining other words, phrases or sentences.**
- **Correlative conjunctions** are pairs of words that connect two parts of a sentence that have an equal value or correlate with each other.

- Examples: **too...to, neither...nor, hardly...when, no sooner...than**, etc.
- We must use the **correct pair** of correlatives.
- The correct pair is '**hardly...when**' and not '**hardly...than**'.
 - **Example:** **Hardly** had I reached the station **when** the train left.
- So, in the given sentence, '**than**' should be replaced with '**when**' to make it grammatically correct.

Correct sentence: *Hardly had he completed his homework **when** the lights went out.*

Additional Information

- When a sentence begins with a **negative word** like '**not**', '**neither**', '**no sooner**', '**hardly**', '**scarcely**', etc., then the sentence will be in **inversion**, i.e., the **helping verb will come before the subject**.

Hinglish

- एक **conjunction** एक शब्द है जो अन्य **words, phrases** या **sentences** को जोड़ने के लिए उपयोग किया जाता है।
- **Correlative conjunctions** ऐसे शब्दों के जोड़े होते हैं जो एक वाक्य के दो हिस्सों को जोड़ते हैं जिनका एक समान मूल्य होता है या एक दूसरे के साथ सहसंबंधित होता है।
 - **उदाहरण:** **too...to, neither...nor, hardly...when, no sooner...than**, आदि।
- हमें सहसंबंधों की सही जोड़ी का उपयोग करना चाहिए।
- सही जोड़ी है 'मुश्किल से ... जब' और 'मुश्किल से ... नहीं'।
 - **उदाहरण:** **Hardly** had I reached the station **when** the train left.
- तो, दिए गए वाक्य में, '**than**' को '**when**' के साथ बदल दिया जाना चाहिए ताकि इसे व्याकरणिक रूप से सही किया जा सके।

Question 54

Identify the word that is opposite in meaning of the word 'INADVERTENT'.

Options:

- A. adequate
- B. available
- C. intentional
- D. negligent

Answer: C

Solution:

The correct antonym is 'Intentional'.

Key Points

- The word 'Inadvertent' means 'done or happening unintentionally (अनजाने में किया या हो रहा है)'.
- The **antonyms** of the word 'Inadvertent' are 'calculated, deliberate, intended, **intentional**, planned, premeditated, premeditative, prepen, set'.
- **From the antonym** of the given word, we can say that the word 'Intentional' is the **opposite in meaning**.
- The word '**Intentional**' means 'planned or intended (नियोजित या इरादा)'.

Hence, the correct answer is **option 3**. Let's see the **meaning** of other given options:-

Words	Meaning
Adequate	enough or satisfactory for a particular purpose (किसी विशेष उद्देश्य के लिए पर्याप्त या संतोषजनक)
Available	able to be bought or used (खरीदा या इस्तेमाल किया जा सकता है)
Negligent	not being careful or giving enough attention to people or things that are your responsibility (सावधान न रहना या लोगों या उन चीज़ों पर पर्याप्त ध्यान न देना जो आपकी ज़िम्मेदारी हैं)

Additional Information

- The **synonyms** of the word 'Inadvertent' are 'accidental, casual, chance, fluky (also flukey), fortuitous, incidental, unintended, unintentional, unplanned, unpremeditated, unwitting'.
- Example of 'Inadvertent' in a sentence:
 - All authors need to be wary of **inadvertent** copying of other people's ideas.

Question 55

Direction: Choose the appropriate word for the given sentence:

Your brother will be failed _____ he does not come tomorrow.

Options:

- A. if
- B. though
- C. unless

D. until

Answer: A

Solution:

The correct answer is 'if.'

Key Points

- We often use **if** to introduce possible or impossible situations or conditions and their results.
- The situations or conditions can be real, imagined, or uncertain.
 - **Example:** *I usually make a sandwich to take to work **if** I have enough time.*
- In this sentence, the use of '**though**', '**unless**', and '**until**' would be incorrect.
 - **Though:** **Though** means '**in spite of something**.'
 - It is a subordinating conjunction.
 - This means that the clause which they introduce is a subordinate clause, which needs a main clause to make it complete.
 - When the **though** clause comes after the main clause, it can also mean '**but it is also true that ...**'
 - **Example:** *We didn't make any profit **though** nobody knows why.*
 - **Unless:** We use the conjunction **unless** to mean '**except if**.'
 - **Example:** *They won't come **unless** you invite them.*
 - **Until:** We use **until** as a subordinating conjunction to connect an action or an event to a point in time.
 - **Example:** *We played chess **until** midnight.*
- Hence, the word, '**if**' should be used to make the sentence grammatically correct.
- Hence, the correct answer is '**if**.'
 - **The Correct Sentence:** *Your brother will be failed **if** he does not come tomorrow.*

Additional Information

- A **conjunction** is a joiner, a word that connects (conjoins) parts of a sentence.
- **Coordinating conjunctions** join equals to one another.
 - **Example:** *Most children like sweets **and** cookies.*
- Coordinating conjunctions go in between items joined, not at the beginning or end.
 - **Example:** *I like coffee, **but** I don't like tea.*

Hinglish

- हम अक्सर संभावित या असंभव स्थितियों या स्थितियों और उनके परिणामों को पेश करने के लिए **if** का उपयोग करते हैं।
- परिस्थितियाँ वास्तविक, काल्पनिक या अनिश्चित हो सकती हैं।
 - **Example:** *I usually make a sandwich to take to work **if** I have enough time.*
- इस वाक्य में '**though**', '**unless**' और '**until**' का प्रयोग गलत होगा।
 - **Though:** **Though** का मतलब है 'कुछ के बावजूद'।
 - यह एक subordinating conjunction है।
 - इसका अर्थ यह है कि वे जिस clause का परिचय देते हैं वह एक subordinate clause है, जिसे पूरा करने के लिए एक main clause की आवश्यकता होती है।
 - जब main clause के बाद **though** उपवाक्य आता है, तो इसका अर्थ यह भी हो सकता है कि 'लेकिन यह भी सत्य है कि...'

- **Example:** *We didn't make any profit **though** nobody knows why.*
 - **Unless:** हम conjunction **unless** का अर्थ 'except if' हैं।
 - **Example:** *They won't come **unless** you invite them.*
 - **Until:** हम **until** किसी क्रिया या घटना को समय पर एक बिंदु से जोड़ने के लिए एक subordinating conjunction के रूप में उपयोग करते हैं।
 - **Example:** *We played chess **until** midnight.*
 - अतः वाक्य को व्याकरणिक रूप से शुद्ध करने के लिए 'if' शब्द का प्रयोग करना चाहिए।
-

Question 56

Select the most appropriate synonym for the given word.

Fury

Options:

- A. Crime
- B. Sin
- C. Anger
- D. Risk

Answer: C

Solution:

The correct answer is **Option 3**.

- **Fury:** wild or violent anger
- **Anger:** a strong feeling of annoyance, displeasure, or hostility.
- **Crime:** an action or omission which constitutes an offence and is punishable by law.
- **Sin:** an immoral act considered to be a transgression against divine law.
- **Risk:** a situation involving exposure to danger.

Anger is synonymous with **fury**.

Question 57

Directions: Choose the appropriate preposition for the given sentence:

His path was beset _____ difficulties.

Options:

- A. with
- B. within
- C. among
- D. by

Answer: A

Solution:

The correct answer is 'with.'



Key Points

- The given sentence is saying something about someone's difficult path.
- The use of the word "beset" in the sentence indicates be covered or studded with.
- Therefore, the most appropriate preposition to be filled in the blank is 'with'.
- 'With' means having or possessing something.
 - Example: The plan was beset **with** difficulties from the beginning.

Hence, the correct answer is option 1.

Complete Sentence: His path was beset **with** difficulties.



Additional Information

- Let us explore the other options:
 - 'Within' means inside something.
 - 'Among' means situated more or less centrally in relation to several other things.
 - 'By' is used for identifying the agent performing an action.

Question 58

What is the antonym of the word given below?

Initiate

Options:

- A. Start
- B. Begin
- C. Commence
- D. Finish

Answer: D

Solution:

The correct answer is **Finish**.



Key Points

Initiate: to cause something to begin

Finish: bring (a task or activity) to an end; complete

Thus, we can see that Initiate and Finish are antonyms.



Additional Information

The meaning of the other words:

- **Start:** to begin doing something
- **Begin:** to start to do something
- **Commence:** to begin something especially a shop or business.
- **Initiate:** to take first step in doing something especially something new.
Start, Begin, Commence and Initiate are all near-synonyms.

Question 59

Select the correctly spelt word.

Options:

- A. Attendance
- B. Attendence
- C. Attandence
- D. Attandance

Answer: A

Solution:

The correct answer is **option 1), i.e. Attendance**.

Key Points

- The correct spelling of the word is '**Attendance**'.
 - The meaning of the word '**Attendance**' is 'the action of being present'. (उपस्थित होने की क्रिया)
-

Question 60

Read the sentence to find out whether there is any error in it. The error, if any, will be in one part of the sentence. The number of that part is the answer. If there is no error, the answer is (5). Ignore errors of punctuation, if any.

(1)How much boys/(2) were present/(3)in the assembly./(4) No Error

Options:

- A. 1
- B. 2
- C. 3
- D. 4

Answer: A

Solution:

Key Points

- Part 1 contains an error.
- We know that in case of uncountable nouns like milk, water, ,rice etc. we use **much** and not **many** but when the uncountable noun is supported by a countable noun like people, packet, bottle, bag etc. we should use **many** and not **much**.
- The question should start as *How many boys*.

Correct Sentence:How many boys were present in the assembly.

Question 61

Identify the segment in the sentence which contains the grammatical error.

We went for swimming into the river on Sunday.

Options:

A. We went for swimming

B. into the

C. river on

D. Sunday

Answer: B

Solution:

Option 2 is the right answer. It has an error.

A preposition is always followed by an object, which is a noun and never a verb. [subject +verb+ preposition+ object]

It joins a noun, a pronoun, or noun phrases to the rest of the sentence **to show their relation**.

'**into**' shows motion. However, in the given sentence, **a location** is indicated ' **a river**'.

The correct preposition would be '**in**', which indicates '**the location of the action**'.

The correct sentence, thus, is --**We went for swimming in the river on Sunday.**

Question 62

Direction: In the following question, out of the four alternatives, select the alternative which will improve the underlined part of the sentence. In case no improvement is needed, select “No improvement”.

No sooner Riya reached the station than she met her friend.

Options:

- A. Did Riya reached
- B. Did Riya reach
- C. Riya did reached
- D. No improvement

Answer: B

Solution:

The correct answer is “**Did Riya reach**”



Key Points

- In the given sentence ‘**no sooner**’ is used at the beginning of the sentence, we used inverted word order, i.e. Auxillary verb comes after no sooner and before the subject. Another example of inverted word order is: 'nosooner did I reach the station than the train had already left'.
- If the sentence starts with the subject and not with no sooner than there is no need to use an auxiliary verb, for example, a given sentence can also be written as “Riya no sooner reached the station than she met her friend.” this is also correct and conveys the same meaning.

Correct sentence: “*No sooner did Riya reach the station than she met her friend.*”

Question 63

Direction: Change the Narration-

“What are you doing?” He said to me

Options:

A.

He asked me what I was doing.

B.

He asked me what I were doing.

C.

He asked me that what I was doing.

D.

He asked me what he was doing.

Answer: A

Solution:

The correct answer is '**He asked me what I was doing.**'



Key Points

- An **interrogative sentence** has a question.
- The rule for changing Direct Speech to Indirect Speech:
 - The given sentence is in **Present Continuous Tense**.
 - **Present Continuous Tense** changes to **Past Continuous tense**.
 - Remove the commas and inverted commas.
 - Do not use **that** in the sentence of indirect speech.
 - The reporting verb **said/said to** is changed in **asked, demanded, ordered, enquired** as per the nature of the sentence.
 - Remove the question mark in the sentence of Indirect speech.
 - There are some questions that cannot be answered using '**YES/NO.**'
 - To make the indirect speech of such questions, '**if/whether**' will not be used.
 - The question is changed to assertive form and is simply placed after the reporting verb.
 - **Example:** *The boy said to Ram, "What is your name?"* (**Direct speech**)
The boy asked Ram what his name was. (**Indirect speech**)
- Therefore, the correct answer is '**He asked me what I was doing.**'



Additional Information

- If the sentence of Direct speech has a question that cannot be answered by saying yes or no then it does not use "**if**" or "**whether**" in its Indirect speech.
 - **Example:** *She said to me, "Where did you buy this dress from?"* (**Direct speech**)
She asked me where I had bought that dress from. (**Indirect speech**)



Hinglish

- **Interrogative** वाक्य में एक प्रश्न होता है।
- Direct Speech को Indirect Speech में बदलने का नियम:
 - दिया गया वाक्य **Present Continuous Tense** में है।
 - **Present Continuous Tense, Past Continuous tense** में बदलता है।
 - Commas और inverted commas हटा दे।
 - Indirect speech के वाक्य में **that** प्रयोग न करें।
 - **Said/said to** रिपोर्टिंग verb वाक्य की प्रकृति के अनुसार **asked, demanded, ordered, enquired** में बदल जाती है।
 - Indirect speech के वाक्य में प्रश्न चिह्न हटा दें।
 - कुछ ऐसे प्रश्न हैं जिनका उत्तर '**YES/NO**' में नहीं दिया जा सकता है।
 - ऐसे प्रश्नों को Indirect speech देने के लिए '**if/whether**' का प्रयोग नहीं किया जाएगा।

- Question को assertive form में बदल दिया जाता है और केवल reporting verb के बाद रखा जाता है।
 - **Example:** *The boy said to Ram, "What is your name?"* (**Direct speech**)
The boy asked Ram what his name was. (**Indirect speech**)
-

Question 64

People in the villages of Rajasthan lead a very simple life. Their way of living has not changed over the years. They live in circular huts. The walls of these huts are covered with cowdung. Every hut has a small place for worship. The life of these people is full of difficulties. It is very hot in summers and cold in winters. Water is a major problem. Sometimes they have to walk a long distance to get drinking water. For their agriculture they depend on rains. But these people are very brave. They have learnt to face difficulties and they never lose hope. They also like to enjoy their life. Women like to wear dresses full of bright colours. People living in villages in Rajasthan have a rich tradition of music and dance. The people of Rajasthan are very proud of their culture.

Which of the following word is a synonym for 'brave'?

Options:

- A. hardworking
- B. valiant
- C. prosperous
- D. enthusiastic

Answer: B

Solution:

A synonym is a word or phrase that means exactly or nearly the same as another word or phrase in the same language.

Meaning of Brave: Courageous, showing no fear, daring

'Valiant' is the synonym of the word 'Brave'.

- **Hardworking:** tending to work with energy and commitment
- **Prosperous:** fortunate financially, flourishing
- **Enthusiastic:** showing great excitement or interest

Hence, a synonym for the word 'Brave' is 'Valiant'.

Question 65

People in the villages of Rajasthan lead a very simple life. Their way of living has not changed over the years. They live in circular huts. The walls of these huts are covered with cowdung. Every hut has a small place for worship. The life of these people is full of difficulties. It is very hot in summers and cold in winters. Water is a major problem. Sometimes they have to walk a long distance to get drinking water. For their agriculture they depend on rains. But these people are very brave. They have learnt to face difficulties and they never lose hope. They also like to enjoy their life. Women like to wear dresses full of bright colours. People living in villages in Rajasthan have a rich tradition of music and dance. The people of Rajasthan are very proud of their culture.

‘The way of living’ can be replaced with the word

Options:

- A. livelihood
- B. liveliness
- C. lifelike
- D. lifestyle

Answer: D

Solution:

‘**The way of living**’ means the particular way that a person lives. It is the way you live including your style, attitudes and possessions.

- **Lifestyle**(noun): The way in which a person lives
- **Livelihood**(noun): The money you need to pay for food, place to live, clothing, etc.
- **Liveliness**(noun): The quality of being interesting and exciting.
- **Lifelike**(adjective): Accurately representing or imitating real life.

Hence, ‘**The way of living**’ can be replaced with the word ‘**Lifestyle**’.

Question 66

In the following question, one part of the sentence may have an error. Find out which part of the sentence has an error and click the button corresponding to it. If the sentence is free from error, click the "No error" option.

If aspirin will ease my headache, I will take a couple tonight.

Options:

- A. If aspirin will ease
- B. No error
- C. I will take a couple tonight.
- D. my headache

Answer: A

Solution:

The correct answer is 'If aspirin will'.



Key Points

- The error lies in **option 1** of the sentence.
 - There will not be two '**will**' in a sentence.
 - If in a clause we use '**will**', '**shall**', '**would**', '**should**', we shall not use it twice.
 - If the subordinate clause is in the future tense ("I *will* take a couple tonight"), then the if clause will be in the present tense ("If aspirin *eases* my headache")
 - The **action** in the **if-clause** is the **aspirin easing the headache**, which will **take place only after** the speaker **takes them later that night**.
 - Hence, the sentence is: "*If aspirin eases my headache, I will take a couple tonight*"
-

Question 67

Select the most appropriate synonym of the given word.

CRUCIAL

Options:

- A. Vital
- B. Easy
- C. Menial

D. Trivial

Answer: A

Solution:

The correct answer is 'Vital'

Key Points

- The word **crucial** means **decisive** or **critical** especially in the success or failure of something.
 - The word **vital** meaning absolutely **necessary** or **essential** is closest to that meaning.
 - Example: *it's absolutely vital to keep a steady oxygen flow for the ICU patient.*
 - Hence, **Vital** is the synonym for **crucial**
-

Question 68

Find the correct spelling in the given option:

Options:

A.

Indepandant

B.

Influentail

C.

Professor

D.

Beneficeal

Answer: C

Solution:

The correct answer is- 'Professor'

Key Points

- Among the given options correct spelling is '**Professor**'.
- The word **Professor** is a noun and its meaning is a **teacher of the highest rank in a department of a British university, or a teacher of high rank in an American university or college.**
- For Example - *She's a **professor** of nutrition at Columbia University.*

Hence the correct answer is **option 3**.

Wrong Spelling	Correct Spelling
Independant	Independent
Influentail	Influential
Beneficeal	Beneficial

Additional Information

- Let's check out some synonyms and antonyms of '**Professor**':
 - **Synonyms**- instructor, teacher, educator
 - **Antonyms**- pupil, student



- दिए गए विकल्पों में सही वर्तनी '**Professor**' है।
- **Professor** शब्द एक **noun** है और इसका अर्थ किसी विश्वविद्यालय या कॉलेज में उच्चतम रैंक का शिक्षक है।
 - For Example: *She's a **professor** of nutrition at Columbia University.*

इसलिए सही उत्तर **option 3** है।

- आइए देखें '**Professor**' के कुछ **Synonyms** और **Antonyms**:
 - **Synonyms**- instructor, teacher, educator
 - **Antonyms**- pupil, student

Question 69

Choose the word which best expresses nearly the same meaning as the word

CONDUCTIVE:

Options:

A. Ineffectual

B. Contributive

C. Prodigious

D. Hindering

Answer: B

Solution:

The correct answer is '**Contributive**'.

Key Points

- The most appropriate synonym of the given word '**Conducive**' is '**Contributive**'.
- **Conducive**:helping or making something happen. (सहायक या किसी का कारण बनना)
 - *Example:This hot weather is not **conducive** to hard work.*
- **Contributive**:helping to cause something. (कुछ पैदा करने में मदद करना)
 - *Example:a **contributive** factor...*

Correct Answer:*Contributive.*

Additional Information

- Let's look at the **meaning** of other words:
 - **Ineffectual**:not having the ability to achieve much; not achieving what you want to. [योग्यता का न होना (कुछ प्राप्त करने के लिए); वांछित सफलता प्राप्त न होना; इच्छित सफलता का न मिलना, निष्प्रभावी; बेकार, व्यर्थ]
 - **Prodigious**:very large or powerful and surprising. (बहुत बड़ा या शक्तिशाली और आश्चर्यजनक)
 - **Hindering**:to make it more difficult for somebody/something to do something. (व्यक्ति या वस्तु के लिए कठिनाई पैदा करना, किसी के काम में बाधा डालना)

Question 70

Select the misspelt word.

Options:

A. Beaten

B. Stolen

C. Writen

D. Eaten

Answer: C

Solution:

The erroneous spelling is "**Writen**"

Key Points

- The incorrect spelling is "Written".
 - Let's see the meanings of the given words-
 - **Beaten**-strike (a person or an animal) repeatedly and violently so as to hurt or injure them, typically with an implement such as a club or whip.
 - **Stolen**-take (another person's property) without permission or legal right and without intending to return it.
 - **Written**-mark (letters, words, or other symbols) on a surface, typically paper, with a pen, pencil, or similar implement.
 - **Eaten**-put (food) into the mouth and chew and swallow it.
 - Hence, the correct answer is **Option 3**.
-